09/06/2026

CODE-P

Aakash
Medical | IIT-JEE | Foundations

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PROJECT PHOENIX FOR RE-NEET-2026

MM: 720 **CST - 07** **Time: 180 Mins.**

Answers

1. (3)	37. (4)	73. (4)	109. (1)	145. (3)
2. (4)	38. (1)	74. (2)	110. (3)	146. (1)
3. (3)	39. (1)	75. (2)	111. (3)	147. (4)
4. (3)	40. (4)	76. (2)	112. (3)	148. (1)
5. (2)	41. (2)	77. (4)	113. (2)	149. (3)
6. (2)	42. (1)	78. (3)	114. (2)	150. (4)
7. (1)	43. (3)	79. (4)	115. (1)	151. (3)
8. (4)	44. (4)	80. (3)	116. (1)	152. (4)
9. (4)	45. (1)	81. (2)	117. (3)	153. (3)
10. (4)	46. (4)	82. (1)	118. (3)	154. (2)
11. (4)	47. (3)	83. (4)	119. (3)	155. (3)
12. (2)	48. (1)	84. (1)	120. (4)	156. (4)
13. (2)	49. (2)	85. (2)	121. (1)	157. (4)
14. (2)	50. (3)	86. (2)	122. (1)	158. (4)
15. (2)	51. (2)	87. (2)	123. (4)	159. (2)
16. (3)	52. (2)	88. (3)	124. (2)	160. (4)
17. (4)	53. (1)	89. (2)	125. (4)	161. (3)
18. (1)	54. (4)	90. (2)	126. (2)	162. (4)
19. (1)	55. (2)	91. (1)	127. (2)	163. (4)
20. (2)	56. (3)	92. (1)	128. (1)	164. (1)
21. (3)	57. (4)	93. (3)	129. (1)	165. (3)
22. (2)	58. (2)	94. (3)	130. (2)	166. (1)
23. (1)	59. (4)	95. (4)	131. (4)	167. (3)
24. (1)	60. (1)	96. (2)	132. (3)	168. (3)
25. (3)	61. (4)	97. (1)	133. (4)	169. (2)
26. (2)	62. (3)	98. (3)	134. (3)	170. (4)
27. (3)	63. (3)	99. (3)	135. (4)	171. (3)
28. (2)	64. (2)	100. (3)	136. (2)	172. (3)
29. (1)	65. (2)	101. (4)	137. (1)	173. (1)
30. (2)	66. (2)	102. (2)	138. (3)	174. (3)
31. (3)	67. (4)	103. (3)	139. (3)	175. (3)
32. (3)	68. (1)	104. (2)	140. (4)	176. (4)
33. (2)	69. (2)	105. (1)	141. (4)	177. (3)
34. (4)	70. (4)	106. (1)	142. (4)	178. (3)
35. (2)	71. (2)	107. (4)	143. (2)	179. (1)
36. (1)	72. (2)	108. (4)	144. (3)	180. (4)

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Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
- It is easier to pull a block as value of friction is less there.
- Action and reaction forces acts simultaneously.

(2) **Answer :** (4)
Solution:
Properties of intrinsic semiconductor.

(3) **Answer :** (3)
Solution:
(1) Due to short wavelength microwaves are used for radar system for aircraft navigation.
(2) Radio waves are mostly used in radio communication such as AM and FM broadcasting.
(3) Infrared rays are used in household remote switches.
(4) Ultraviolet rays are used in water purifiers.

(4) **Answer :** (3)
Solution:
Let θ be temperature of interface, then in series heat current must be same.
i.e. $\frac{\Delta\theta_1}{R_1} = \frac{\Delta\theta_2}{R_2}$, $R = \frac{l}{KA}$
⇒ $\frac{2KA}{l}(100-\theta) = \frac{KA}{l}(\theta - 40)$
⇒ 200 – 2θ = θ – 40
240 = 3θ
⇒ $\boxed{\theta = 80^\circ C}$

(5) **Answer :** (2)
Solution:
$\vec{L} = \vec{r} \times \vec{p}$
$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k})$ m , $\vec{p} = (10\hat{j})$ kg m s^{-1}

$$\vec{L} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 0 & 10 & 0 \end{vmatrix}$$

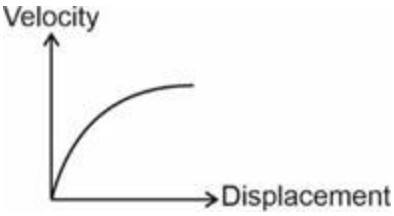
⇒ $\hat{i}\,(0-10) - \hat{j}\,(0-0) + \hat{k}\,(10-0)$
= $(-10\hat{i} + 10\hat{k})$ kg m^2 s^{-1}

(6) **Answer :** (2)
Solution:
$x = \frac{mL}{M+m}$
= $\frac{60(10)}{360} = \frac{5}{3} = 1.67$ m

(7) **Answer :** (1)
Solution:
$x = t^3 + t^2$
$v = 3t^2 + 2t$,
$v\,(t = 2\text{ s}) = 3(2)^2 + 2(2) = 16$ m/s
$a = 6t + 2$
$a\,(t = 2\text{ s}) = 14$ m/s^2
$P = \vec{F} . \vec{v} = M\,a \times v$
= 10 × 14 × 16 = 2240 W

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(8) **Answer :** (4)
Solution:
$a = \frac{vdv}{ds}$
If *a* = constant, then
$\int_0^s ads = \int_0^v vdv \Rightarrow as = \frac{v^2}{2} \Rightarrow v^2 = 2as$



(9) **Answer :** (4)
Solution:
$\vec{A} \times \vec{B}$ is anti-parallel to $\vec{B} \times \vec{A}$
⇒ α = 180°
$\vec{A} \times \vec{B}$ is perpendicular to $\vec{A}$
⇒ $\beta = 90^\circ$
∴ α + β = 270°

(10) **Answer :** (4)
Solution:
$MK^2 = \frac{MR^2}{4}$
= moment of inertia about diameter.
⇒ $K^2 = \frac{R^2}{4} \Rightarrow$ Radius of gyration(K) = $\frac{R}{2}$

(11) **Answer :** (4)
Solution:
Since, the particle is in uniform circular motion, its angular acceleration is zero.

(12) **Answer :** (2)
Solution:
Yes the value of potential energy depends upon the reference taken but change in potential energy is independent of reference level.

(13) **Answer :** (2)
Solution:
$\because a_m = \frac{\int a_1 dt + \int a_2 dt}{(t_1 + t_2)}$
$a_m = \frac{a_1 t_1 + a_2 t_2}{t_1 + t_2}$
$a_m = \frac{a_0 T + \frac{2a_0 T}{2}}{T + \frac{T}{2}} = \frac{4a_0}{3}$

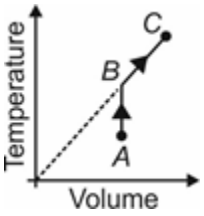
(14) **Answer :** (2)
Solution:
$A_1 V_1 = A_2 V_2 + A_3 V_3$
$4A = 0.4A \times 2 + \frac{AV}{2}$
$4 = 0.8 + \frac{V}{2}$
$3.2 = \frac{V}{2}$ ⇒ V = 6.4 m/s

(15) **Answer :** (2)
Solution:
$y = 0.4 \sin\left(2\pi t + \frac{\pi}{3}\right)$
For initial phase put *t* = 0
we get $\frac{\pi}{3}$ as initial phase

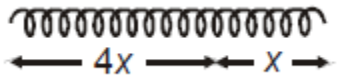
(16) **Answer :** (3)

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Hint:
Isochoric and isobaric process.
Solution:
$A \rightarrow B$ is isochoric process, pressure increases therefore temperature will increase.
In temperature-volume curve line will be parallel to temperature axis
$B \rightarrow C$ is isobaric process, volume is increasing, therefore temperature will also increase by
$P_0V = nRT$
$V \propto T$



(17) **Answer :** (4)
Solution:
Spring constant $\propto \frac{1}{\text{length}}$



$k = \frac{b}{5x}$; $k' = \frac{b}{x}$; $\therefore \quad k' = 5\text{k}$

(18) **Answer :** (1)
Solution:
Answer (1)
PV = constant
PdV + VdP = 0
$B = -\frac{dP}{\frac{dV}{V}} = P$

(19) **Answer :** (1)
Hint:
Beat frequency = $f_1 \sim f_2$
Solution:
$f_1 = \frac{400\pi}{2\pi} = 200$ Hz
$f_2 = \frac{398\pi}{2\pi} = 199$ Hz
Beat frequency = 1 Hz
Beats per minute = 60

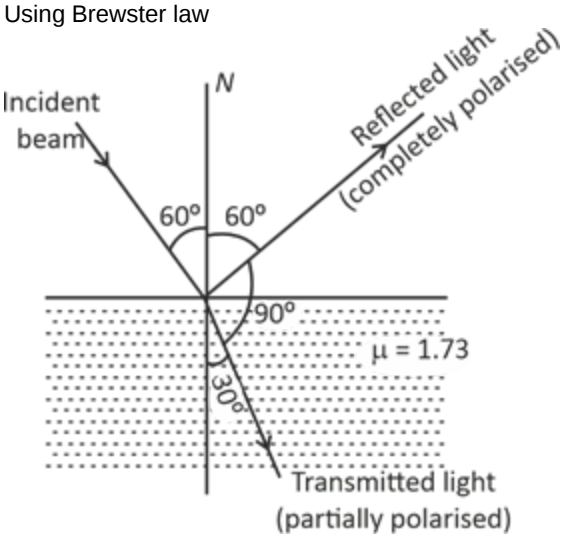
(20) **Answer :** (2)
Solution:
$F = m\omega^2 R = 3.6 \times 10^{22}$ N.

(21) **Answer :** (3)
Solution:
$[Rch] = \left[\frac{hc}{\lambda}\right] = \left[ML^2T^{-2}\right]$

(22) **Answer :** (2)
Hint:
Use the concept of potential due to hollow sphere.
Solution:
Let us assume that charge q appears at inner sphere.
$\Rightarrow \; V_{inner\ sphere} = \frac{K \cdot q}{a} + \frac{K(-Q)}{b} = 0$
$\Rightarrow \; \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{a} - \frac{Q}{b}\right] = 0$
$\Rightarrow \; \frac{q}{a} = \frac{Q}{b}$
$\Rightarrow \; q = \frac{Qa}{b}$

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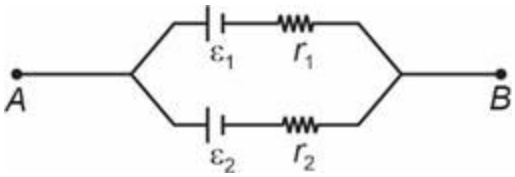
(23) Answer : (1)
Solution:
Using Brewster law



$\mu = \tan\theta_P$
$\Rightarrow 1.73 = \tan\theta_P$
$\Rightarrow \quad \sqrt{3} = \tan\theta_P$
$\Rightarrow \theta_P = 60^\circ$
At this polarising angle, reflected light is perfectly polarized and transmitted light is partially polarised.

(24) Answer : (1)
Solution:
Charge on the plates of a disconnected capacitor remains same.
Capacitance $\Rightarrow C = \frac{A\varepsilon_0}{d}$, if $d \downarrow$ then $C \uparrow$
Energy stored $\Rightarrow U = \frac{Q^2}{2C}, C \uparrow U \downarrow$
Potential difference $\Rightarrow Q = CV,\ C \uparrow,\ V \downarrow$

(25) Answer : (3)
Solution:



$\varepsilon_{AB} = \frac{\frac{\varepsilon_1}{r_1}+\frac{\varepsilon_2}{r_2}}{\frac{1}{r_1}+\frac{1}{r_2}}$ and $r_{eff} = \frac{r_1 r_2}{r_1+r_2}$

(26) Answer : (2)
Solution:
$L = 2\pi R$
$R = \frac{L}{2\pi}$
&
$M = IA$
$= I\pi R^2 = I\pi\left(\frac{L}{2\pi}\right)^2 = \frac{IL^2}{4\pi}$

(27) Answer : (3)
Solution:
$\frac{1}{f} = (\mu_g - 1)\ \frac{2}{R} = \frac{1}{R}, \left(\mu_g = \frac{3}{2}\right), R = f$
$\frac{1}{f_1} = -(\mu_W - 1)\ \frac{2}{R} = \frac{-2}{3R} = \frac{-2}{3f}$
$\frac{1}{f_{eq}} = \frac{1}{f} + \frac{1}{f} + \frac{1}{f_1} = \frac{1}{R} + \frac{1}{R} - \frac{2}{3R}$

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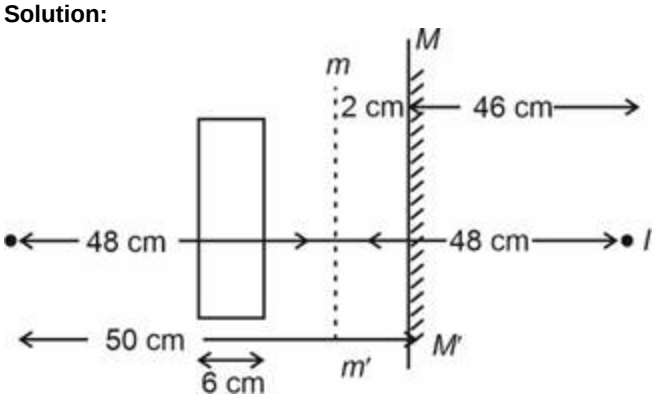
$\frac{1}{f_{eq}} = \frac{1}{f} + \frac{1}{f} = \frac{2}{3f}$

$\frac{1}{f_{eq}} = \frac{2}{f} - \frac{2}{3f}$

$\frac{1}{f_{eq}} = \frac{4}{3f}$

$f_{eq} = \frac{3f}{4}$

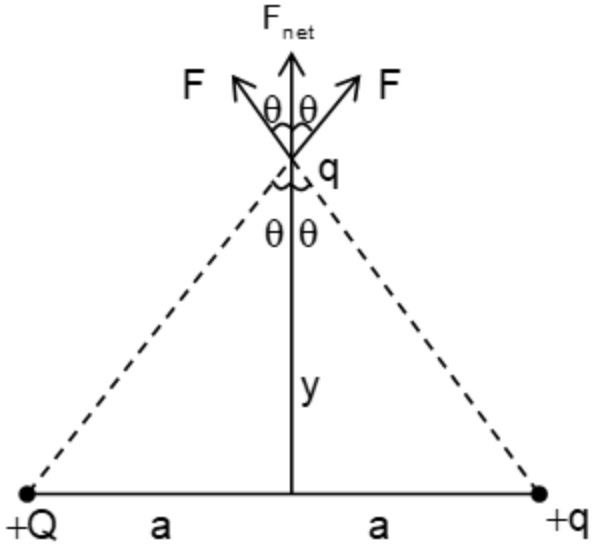
(28) Answer : (2)
Solution:



Shift due to slab = $6\left[1 - \frac{2}{3}\right] = 2 \text{ cm}$

So glass shift mirror from *M'* to *mm'* distance of object from this virtual mirror will be = 50 – *x* = 50 – 2 = 48 cm
The virtual mirror will form image of object *O* at a distance 48 cm behind it, so distance of image from actual mirror *MM'* is = 48 – 2 = 46 cm
Distance of image as seen by observer = 50 + 46 = 96 cm

(29) Answer : (1)
Solution:



$F_{net} = 2F\cos\theta$

$= 2\frac{Qq}{4\pi\varepsilon_0(a^2+y^2)}\frac{y}{\sqrt{(a^2+y^2)}}$

$= \frac{2Qqv}{4\pi\varepsilon_0(a^2+y^2)^{3/2}}$

For maximum value of force
$\frac{dF}{dy} = 0 \Rightarrow y = \frac{a}{\sqrt{2}}$

(30) Answer : (2)
Solution:

$\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

(31) Answer : (3)
Solution:
In the given circuit
Impedance, $Z = \sqrt{(30)^2 + (80-40)^2}$
$Z = \sqrt{30^2 + 40^2}$
...[1 Mark]
Z = 50 Ω
$I_0 = \frac{E_0}{Z} = \frac{200\sqrt{2}}{50}$
$I_0 = 4\sqrt{2}$ A
...[1 Mark]

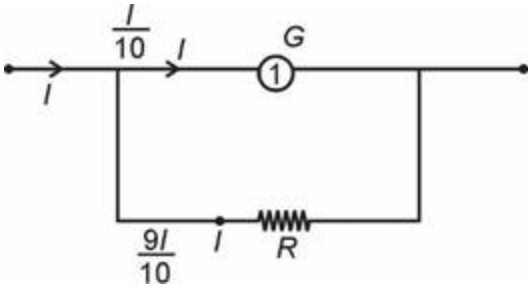
(32) Answer : (3)
Solution:
$\vec{J} = \sigma \vec{E}$
Current density $\vec{J}$ is directed along $\vec{E}$.
So angle between $\vec{J}$ and $\vec{E}$ is 0°.

(33) Answer : (2)
Solution:
$L = \frac{nh}{2\pi}$
By correspondance with $L = \frac{2h}{\pi}$
n = 4
T.E. = $\frac{-13.6\text{ eV}}{n^2} = \frac{-13.6\text{ eV}}{(4)^2}$
T.E. = –0.85 eV
P.E. = 2(T.E) = –1.7 eV

(34) Answer : (4)
Solution:
The cut-in voltage of a germanium diode is approximately is 0.3 V

(35) Answer : (2)
Solution:
$r = \frac{mv}{qB} = \frac{\sqrt{2mE_k}}{qB} = \frac{\sqrt{2mqV}}{qB}$
For same potential and same charge
$r \propto \sqrt{m}$
$m \propto r^2$
$\frac{m_x}{m_y} = \frac{a^2}{b^2}$

(36) Answer : (1)
Solution:



$\frac{I}{10} \times G = \frac{9}{10} \times R$
$R = \frac{G}{9} = \frac{45}{9} = 5\ \Omega$

(37) Answer : (4)

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Solution:
For all value $Y = 1$

(38) **Answer :** (1)
Solution:
μ_r is greater than 1 for paramagnetic substance
and $\chi_m = \mu_r - 1$
$\chi_m > 0$

(39) **Answer :** (1)
Solution:
Radius of nucleus (R) = $R_0(A)^{1/3}$
$\Rightarrow 3.6 \times 10^{-15} = 1.2 \times 10^{-15} (A)^{1/3}$
$\Rightarrow (A)^{1/3} = 3$
$\Rightarrow \boxed{A = 27}$

(40) **Answer :** (4)
Solution:
Magnetic field due to a long straight wire at a distance r is given by
$B = \frac{\mu_0 i}{2\pi r}$
$= \frac{4\pi \times 10^{-7} \times 8}{2\pi \times 2 \times 10^{-2}}$
$= 8 \times 10^{-5}$ T
= 80 μT

(41) **Answer :** (2)
Solution:
At depth d from the surface of earth
$g' = g\left(1 - \frac{d}{R}\right)$
$\Rightarrow g' = \frac{50}{100} g = \frac{g}{2}$
So, $\frac{g}{2} = g\left(1 - \frac{d}{R}\right)$
On solving $d = \frac{R}{2}$

(42) **Answer :** (1)
Solution:
Self inductance is inertia of electricity. Self inductance and mutual inductance have same unit.

(43) **Answer :** (3)
Solution:
$\left[\overrightarrow{\mathrm{L}}\right] = \left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]$

(44) **Answer :** (4)
Solution:
$\overrightarrow{v}_A = \left(\hat{i} + \hat{j}\right)$ m/s
$\overrightarrow{v}_B = \left(4\hat{i} - 3\hat{j}\right)$ m/s
$\overrightarrow{v}_{B/A} = \overrightarrow{v}_B - \overrightarrow{v}_A$
$\overrightarrow{v}_{B/A} = \left(4\hat{i} - 3\hat{j}\right) - \left(\hat{i} + \hat{j}\right)$
$\overrightarrow{v}_{B/A} = \left(3\hat{i} - 4\hat{j}\right)$
$\left|\overrightarrow{v}_{B/A}\right| = 5$ m/s

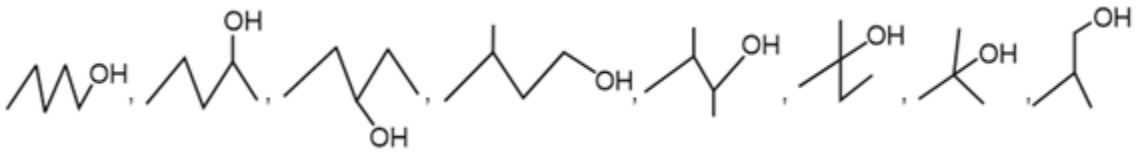
(45) **Answer :** (1)
Hint:
$KE_{max} = eV_0 = h\nu - \phi$
Solution:
$eV_{0_1} = h\nu_1 - \phi$

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$eV_{0_2} = h\nu_2 - \phi$
$V_{0_1} > V_{0_2} \Rightarrow \nu_1 > \nu_2$

CHEMISTRY

(46) **Answer :** (4)
Solution:



(47) **Answer :** (3)
Solution:

$$BaCl_2 + Al_2(SO_4)_3 \longrightarrow BaSO_4 + 2AlCl_3$$

Number of milli equivalence
20 × 0.1 = 2 0.2 × 30 = 6
0 4 m. eq. left
∴ meq. of $BaSO_4$ formed = 2
$\frac{W}{E} \times 1000 = 2$
$W = \frac{2 \times E}{1000} = \frac{2 \times 233}{2 \times 1000} = 0.233\, g$

(48) **Answer :** (1)
Solution:
For H atom; Z = 1
For Li atom, Z = 3
Hence, $E_{2S}(H) > E_{2S}(Li)$

(49) **Answer :** (2)
Solution:
• Electrons are not filled according to (n + l) rule, so, Aufbau rule is violated.
• Electronic configuration of 4p violates the Hund's rule of maximum multiplicity.

(50) **Answer :** (3)
Hint:
Oxides of alkali and alkaline earth metals are basic in nature.
Solution:
MgO : Basic
CO : Neutral
Al_2O_3 : Amphoteric
Cl_2O_7 : Acidic

(51) **Answer :** (2)
Solution:

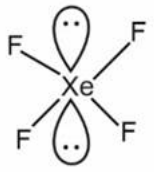
$$XeF_4 + O_2F_2 \xrightarrow{143\ K} \underset{(X)}{XeF_6} + \underset{(Y)}{O_2}$$

$$XeF_4 + 12H_2O \longrightarrow 4Xe + \underset{(Z)}{2XeO_3} + 24\ HF + \underset{(Y)}{3O_2}$$

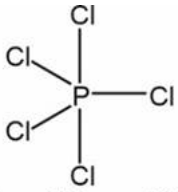
$$XeF_6 + 3H_2O \longrightarrow \underset{(Z)}{XeO_3} + 6HF$$

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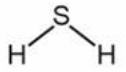
(52) Answer : (2)
Solution:



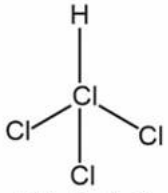
(Square planar)
planar and non polar



(Trigonal bipyramidal)
Non-planar and non polar



(Bent)
Planar and polar



(Tetrahedral)
Non-planar and polar

(53) Answer : (1)
Solution:
H_2^- is less stable than H_2^+ because in $H_2^{\ominus}$ electron enters in antibonding (σ*1s) molecular orbital.
$\because$ Order of BDE is $H_2^- < H_2^+ < H_2$

(54) Answer : (4)
Solution:
$W = -P_{ext}\Delta V = -4 \times (3.5 - 2.5)$
= –4 L atm
= –405.2 J

(55) Answer : (2)
Solution:
- For adiabatic reversible expansion process
  $q = 0$
  $w < 0$
  Hence $\Delta U < 0$
  $\Delta T < 0$
- Work done is non zero for cyclic process
- On stretching of rubber entropy decreases due to uncoiling of molecules.

(56) Answer : (3)
Solution:
$FeO(s) + CO(g) \rightleftharpoons Fe(s) + CO_2(g)$
$K_p = \frac{P_{CO_2}}{P_{CO}} = \frac{x}{3-x} = 0.5$
2x = 3 – x ⇒ x = 1 atm

(57) Answer : (4)
Hint:
Salt of weak acid and weak base undergo both cationic and anionic hydrolysis
Solution:
1. $(NH_4)_2SO_4$ – Salt of weak base and strong acid so, NH_4^+ will undergo cationic hydrolysis
2. CH_3COONa – Salt of weak acid and strong base so CH_3COO^- will undergo anionic hydrolysis.
3. CH_3COONH_4 – Salt of weak acid and weak base, so, both CH_3COO^- and NH_4^+ will undergo hydrolysis i.e., anionic and cationic hydrolysis.
4. Na_2SO_4 – Salt of strong acid and strong base and it will not undergo hydrolysis.

(58) Answer : (2)
Hint:
Oxidising agent are required for oxidation process.
Solution:
$2Cl^- \rightarrow Cl_2$ (oxidation)
$SO_2 \rightarrow SO_3$ (oxidation)

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$Cr_2O_7^{2-}$ → Cr^{3+} (reduction)
$SnCl_4$ → $SnCl_2$ (reduction)

(59) **Answer :** (4)
Solution:

$$Fe^{2+} + Cr_2O_7^{2-} \rightarrow Fe^{3+} + 2\,Cr^{3+}$$

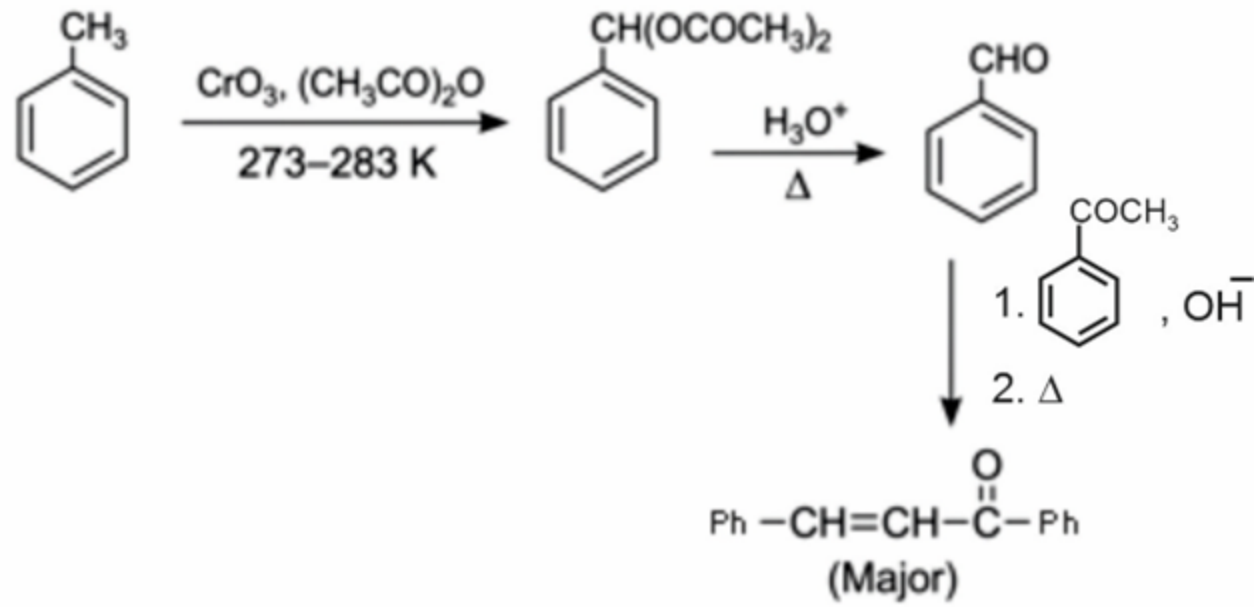
$$6\,Fe^{2+} + Cr_2O_7^{2-} \rightarrow 6\,Fe^{3+} + 2\,Cr^{3+}$$

$$6\,Fe^{2+} + Cr_2O_7^{2-} + 14H^{+} \rightarrow 6\,Fe^{3+} + 2\,Cr^{3+} + 7H_2O$$

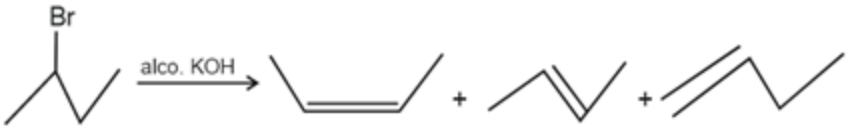
$x = 6,\ y = 1,\ z = 14$

(60) **Answer :** (1)
Hint:
Boron is a 2nd period member
Solution:
Boron is unable to form BF_6^{3-} ion due to unavailability of *d*-orbitals, it cannot expand its octet.
Boron cannot exceed maximum covalence of 4.

(61) **Answer :** (4)
Solution:



(62) **Answer :** (3)
Solution:



(63) **Answer :** (3)
Solution:
1 mole of NO_2 will occupy 22.4 L at STP.

$$\text{Density} = \frac{\text{Molar mass}}{\text{Molar volume}} = \frac{46}{22.4} = 2.05 \text{ g/L}$$

(64) **Answer :** (2)
Solution:
Higher the molar mass, higher is the boiling points. For isomeric alkanes, more is the branching, lesser is the surface area, lesser is the van der Waals force of attraction and lesser is the boiling point.

(65) **Answer :** (2)
Solution:

$$2\,CH_4 + O_2 \xrightarrow[100\ atm]{Cu,\ 523\,K} 2\,CH_3\,OH$$

$$CH_4 + H_2O \xrightarrow[\Delta]{Ni} CO + 3H_2$$

$$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2\,O_3} HCHO + H_2O$$

$$2\,CH_3\,CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3\,COO)_2\,Mn} 2\,CH_3\,COOH + 2H_2O$$

(66) **Answer :** (2)
Solution:
In case of positive deviation from Raoult's law, A-B interactions are weaker than those between A-A or B-B interactions.

(67) **Answer :** (4)
Solution:
$\pi_{Na_2\,SO_4} = \pi_{glucose}$
$i_1 c_1 = i_2 c_2, \quad \begin{pmatrix} i_1 = 1 + 2\alpha \\ i_2 = 1 \end{pmatrix}$
0.004 (1 + 2α) = 0.0108
1 + 2α = 2.7
α = 0.85
% Dissociation = α × 100
= 85%

(68) **Answer :** (1)
Hint:
1 Faraday charge displaces 1 equivalent substance.
Solution:
Equivalent of O_2 = $\frac{5600}{5600} = 1$ eqv.
1 eqv. Of O_2 = 0.25 mole
Charge required for 5600 mL of O_2 gas = 1 Faraday

(69) **Answer :** (2)
Solution:
$2H^+(aq) + 2e^- \rightarrow H_2(g)$
$E_{H^+/H_2} = 0 - \frac{0.0591}{2}\log\left[\frac{p_{H_2}}{[H^+]^2}\right] = -\frac{0.0591}{2}\log\left[\frac{1}{0.1^2}\right] = -\frac{0.0591}{2}\log 100$
= – 0.0591 volt

(70) **Answer :** (4)
Hint:
Order of the reaction may not be equal to the number of molecule of reactant as seen from balanced equation.
Solution:
Order of the reaction may not be equal to the number of molecule of reactant as seen from balanced equation.

(71) **Answer :** (2)
Solution:
Temperature coefficient is 2.
$\therefore \quad r = 2^6 = 64$

(72) **Answer :** (2)
Solution:

Compounds	Melting point (K)
H_2Te	222
H_2S	188
H_2O	273

(73) **Answer :** (4)
Solution:

Molecules	Bond dissociation enthalpy (kJ/mol^{-1})

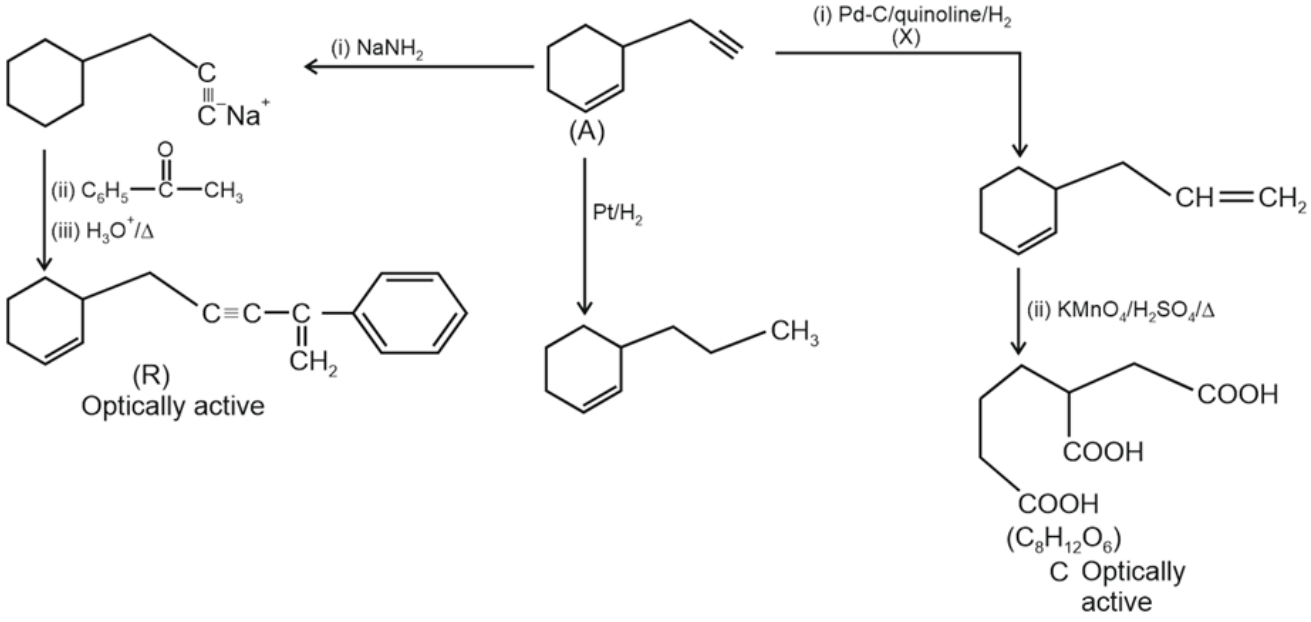
F_2	158.8
Cl_2	242.6
Br_2	192.8
I_2	151.1

(74) **Answer :** (2)
Solution:

$$MnO_2 \xrightarrow[\text{oxidised with air}]{\text{Fused with KOH}} \underset{(A)}{MnO_4^{2-}} \xrightarrow[\text{in alkaline solution}]{\text{Electrolytic oxidation}} \underset{(B)}{MnO_4^-}$$

(75) **Answer :** (2)
Hint:
Interstitial compounds are those which are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.
Solution:
They have high melting points, higher than those of pure metals.

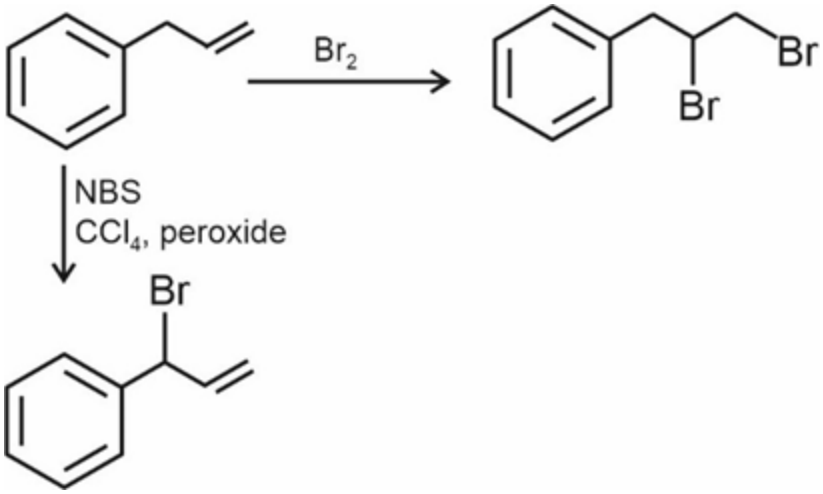
(76) **Answer :** (2)
Solution:



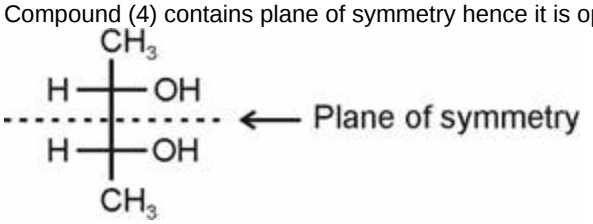
(77) **Answer :** (4)
Hint:
Ionisation isomerism arises when the counter ion in a complex salt is itself a potential ligand and can displace a ligand which can then become the counter ion.
Solution:

Complex	**Magnetic character**	**Isomerism**
$[Cr(NH_3)_4Br_2]Br$	Cr^{3+} is d^3, hence paramagnetic	Cis-trans
$[Ti(H_2O)_5Br](NO_3)_2$	Ti^{3+} is d^1, hence paramagnetic	Ionization
$[Pt(en)(NH_3)Br]NO_3$	Pt^{2+} is d^8, complex is square planar, all electrons are paired	Ionization
$[Co(NH_3)_4(NO_3)_2]NO_3$	Co^{3+} is d^6, hence diamagnetic	Cis-trans

(78) **Answer :** (3)
Solution:
When alkyl-substituted arenes are treated with NBS in presence of CCl_4 and peroxides, benzyl bromides are mainly product

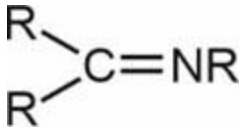


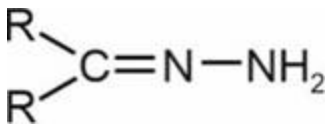
(79) Answer : (4)
Hint:
The compound which contains plane of symmetry or centre of symmetry will be optically inactive.
Solution:
Compound (4) contains plane of symmetry hence it is optically inactive.

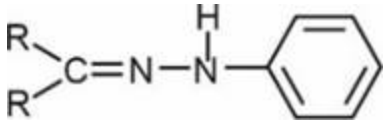


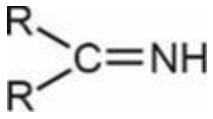
(80) Answer : (3)
Solution:
:CCl_2, dichloro carbon is intermediate

(81) Answer : (2)
Solution:

 - Shift base

 - Hydrazone

 - Phenyl hydrazone

 - Imine

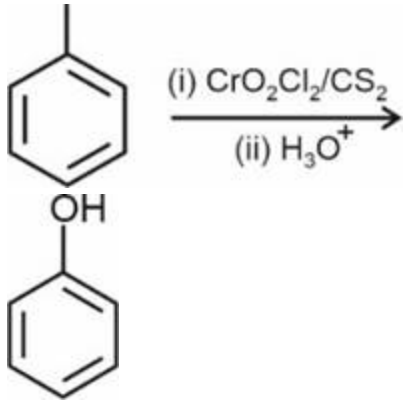
(82) Answer : (1)
Solution:

$$RCN \xrightarrow[(ii)\ H_3O^+]{(i)\ SnCl_2/HCl} RCHO$$

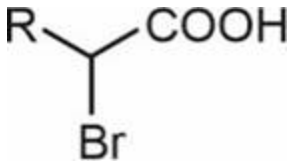
Stephen reaction

Etard reaction

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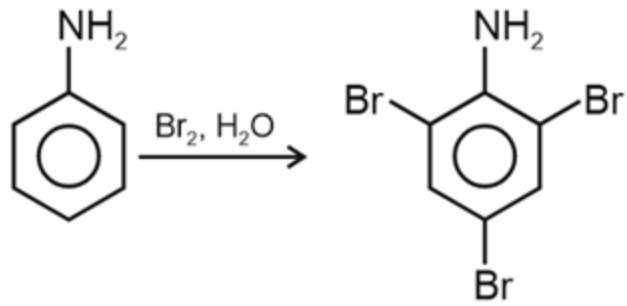


$HCHO \xrightarrow{OH^-} CH_3OH + HCOO^-$ Cannizzaro reaction

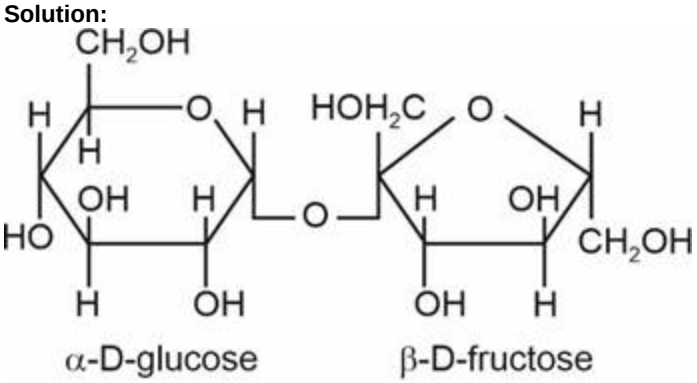
$RCH_2COOH \xrightarrow{Red\ P/Br_2}$  Hell-Volhard-Zelinsky reaction

(83) Answer : (4)
Solution:
Benzaldehyde is less reactive than formaldehyde towards nucleophilic addition reaction.

(84) Answer : (1)
Solution:



(85) Answer : (2)
Hint:
Sucrose is a non-reducing sugar
Solution:

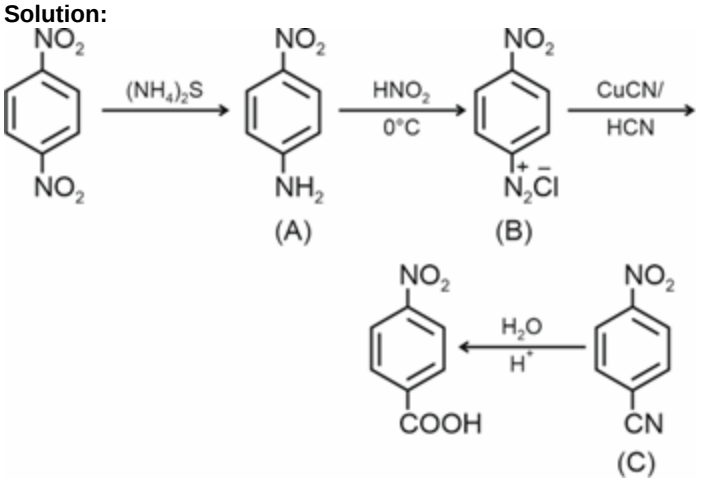


Sucrose

- Sucrose is composed of α-D-glucose and β-D-fructose units
- It is a non-reducing sugar, hence does not reduce Tollens' reagent.

(86) **Answer :** (2)
Solution:
During denaturation of proteins, secondary and tertiary structure are destroyed but primary structure remains intact.

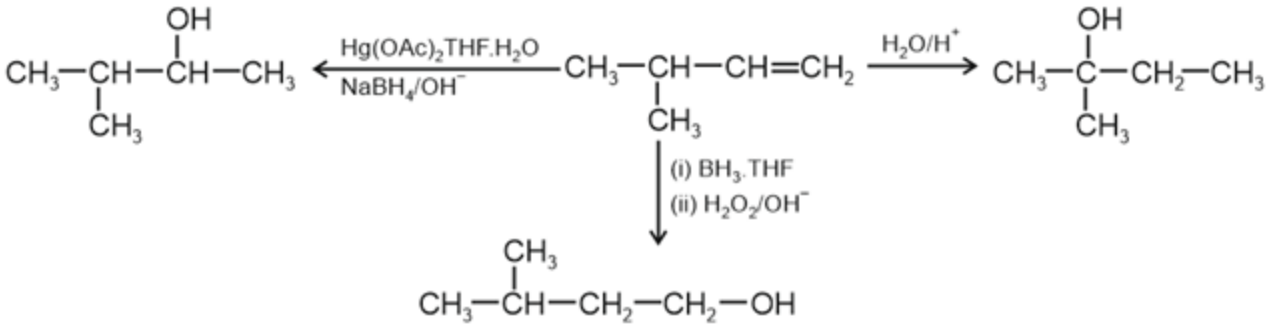
(87) **Answer :** (2)
Solution:



(88) **Answer :** (3)
Solution:
Picric acid is stronger acid due to low pK_a.

(89) **Answer :** (2)
Solution:
Starch paper turns yellow when comes in contact with bromine vapours.

(90) **Answer :** (2)
Solution:



BIOLOGY

(91) **Answer :** (1)
Solution:
Musca is genus of housefly.

(92) **Answer :** (1)
Solution:
Neurospora crassa is often employed in biochemical and genetic work.

(93) **Answer :** (3)
Solution:
Slime moulds produce fruiting bodies during unfavourable conditions because the spores have extremely resistant true walls and survive for many year.

(94) **Answer :** (3)
Solution:

Salvinia being a heterosporous pteridophyte, shows event precursor to seed habit. Moss, like *Funaria*, shows elaborate mechanism of spore dispersal. In gymnosperms, male and female gametophytes do not have independent free-living existence.

(95) **Answer :** (4)
Solution:
Red algae have chlorophyll *a*, cellulosic cell wall and specialised sex organs. They lack flagella throughout their life cycle.

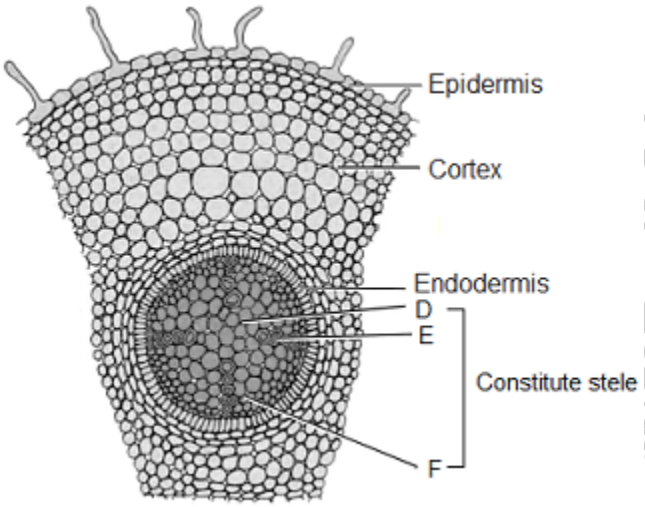
(96) **Answer :** (2)
Solution:
Bentham and Hooker's classification system was a natural classification system. Artificial classification system was given by Linnaeus.

(97) **Answer :** (1)
Solution:
In roots, the root hairs arise from zone of maturation. This zone is differentiated zone thus bearing root hairs.

(98) **Answer :** (3)
Solution:
In monocotyledons, leaf base expands into a sheath covering the stem partially or wholly.

(99) **Answer :** (3)
Solution:
Endodermis of stem may contain starch grains. This is the reason endodermis is also called starch sheath.

(100) **Answer :** (3)
Solution:



(101) **Answer :** (4)
Hint:
Schwann proposed hypothesis that the bodies of animals and plants are composed of cell and product of cell.
Solution:
Theodore Schwann studied different type of animal cells and reported that cell had a thin outer layer.

(102) **Answer :** (2)
Solution:
Detoxification of drugs is carried out by smooth endoplasmic reticulum.

(103) **Answer :** (3)
Hint:
As the H^+ ion concentration increases, pH value decreases.
Solution:
Lysosomes contain enzymes which are capable of digesting carbohydrates, proteins, lipids and nucleic acids. Acidic conditions are maintained inside the lysosome by pumping H^+ ions into them.

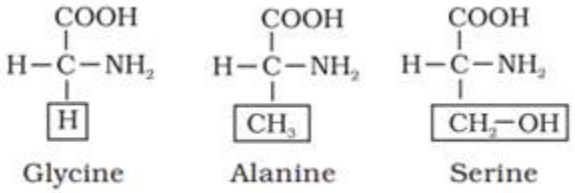
(104) **Answer :** (2)
Hint:
Endocytosis is intake of materials in the form of carrier vesicles formed by invagination of small regions of plasma membrane.
Solution:
Functions like cell growth, formation of intercellular junctions, secretion etc. are due to fluid nature of plasma membrane.

(105) Answer : (1)
Solution:
Genetic variability increases due to meiotic crossing over.

(106) Answer : (1)
Solution:
Skeletal muscle fibres are striated and voluntary whereas cardiac muscle fibres are striated and involuntary.

(107) Answer : (4)
Solution:
Brush-bordered cuboidal epithelium is present in PCT. Cuboidal epithelium has cube-like cells with round nuclei that are present in the centre of the cells.
Basement membrane is avascular.
Basal nuclei is present in columnar epithelium.
Squamous epithelium has flattened cells with irregular boundaries.

(108) Answer : (4)
Solution:



Amino acids

(109) Answer : (1)
Solution:
Conjugated enzyme is known as holoenzyme which is made up of apoenzyme and coenzyme.

(110) Answer : (3)
Hint:
Pancreatic enzyme
Solution:
(a) Trypsin – Enzyme
(b) GLUT-4 – Enables glucose transport into cells
(c) Polysaccharide – Glycosidic bond
(d) Antibody – Fights infectious agents

(111) Answer : (3)
Solution:
Some amount of air always remains in lungs because the sub-atmospheric intrapleural pressure keeps alveoli slightly inflated.

(112) Answer : (3)
Hint:
Amphibians and reptiles have 3-chambered heart.
Solution:
In reptiles, the left atrium receives oxygenated blood from the lungs.
During a cardiac cycle, the blood pumped out by each ventricle is approximately 70 mL, known as the stroke volume.
Neural signals through the sympathetic nerves can increase the rate of heart beat, the strength of ventricular contraction and thereby the cardiac output.

(113) Answer : (2)
Solution:
Hint : Co-enzymes are organic compounds.
Sol. : Zinc acts as a cofactor for enzyme carbonic anhydrase but it is not a coenzyme. Coenzymes are organic compounds like NAD. Ionizable nature of amino and carboxylic groups produces zwitter ions.

(114) Answer : (2)
Hint:
Release of CO_2 is the last step.
Solution:
Respiration involves the following steps:
• Atmospheric air is drawn in and CO_2 rich alveolar air is released out. (Pulmonary ventilation)
• Diffusion of gases across alveolar membrane.
• Transport of gases by the blood.

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- Diffusion of gases between blood and tissues.
- Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2. (Cellular respiration)

(115) Answer : (1)

Hint:

Incomplete double circulation

Solution:

In amphibians and reptiles, the left atrium receives oxygenated blood from gills / lungs / skin and the right atrium gets the deoxygenated blood from other body parts. However, they get mixed up in single ventricle which pumps out mixed blood.

(116) Answer : (1)

Solution:

The correct sequence of events in meiosis-I are as follows:

DNA replication begins in the structure which was first described by Scientist Robert Brown. (S-phase)

↓

The chromosome material become untangled during the process of chromatin condensation. (Leptotene)

↓

The two chromosome which are similar in form, size and structure come to lie side by side in pairs. (Zygotene)

↓

X shaped structure move or shift to the tip of chromosome and gradually disappear. (Diakinesis)

↓

The chromosome reach the poles and spindle fibre completely disappear. (Telophase-I)

(117) Answer : (3)

Solution:

Cells which do not divide further exit G_1 phase to enter an inactive stage where cells remain metabolically active and no longer proliferate unless called on to do so.

(118) Answer : (3)

Solution:

The oxygenase activity of RuBisCO enzyme in C_3 plants leads to the formation of 1 molecule of 3C compound and 1 molecule of 2C compound.

(119) Answer : (3)

Solution:

ATP synthase (CF_1) is present towards the stroma side of chloroplast.

(120) Answer : (4)

Solution:

During the process that occurs in ETS of mitochondria, ubiquinone receives electrons from NADH and $FADH_2$.

(121) Answer : (1)

Solution:

Tripalmitin is a fatty acid.

The ascending order of the substrate with respect to their RQ value is as follows:

Tripalmitin acid < Proteins < Glucose < Organic acids

(122) Answer : (1)

Solution:

Absolute growth rate of A and B = 7 cm^2

Relative growth rate of A and B is 100% and 10% respectively

(123) Answer : (4)

Solution:

Coconut milk can be used to counteract apical dominance, as it is a source of cytokinins.

(124) Answer : (2)

Solution:

Water lily is pollinated by insects.

(125) Answer : (4)

Solution:

There is no need of emasculation when plants produce unisexual flowers.

(126) Answer : (2)

Solution:

The correct order of post-pollination events is as follows:

1. (C) Pistil accepts the pollen and promotes the post-pollination events.

2. (A) The contents of the pollen grain move into the pollen tube.
3. (B) Pollen tube enters one of the synergids through the filiform apparatus.
4. (D) Pollen tube discharges its cytoplasmic content.
5. (E) Gametes fuse with secondary nucleus and egg cell.

The correct sequence will be (C) → (A) → (B) → (D) → (E)

(127) Answer : (2)
Solution:
Life cycle of *Drosophila* is of two weeks.
Male and female are easily distinguishable.

(128) Answer : (1)
Solution:
In AaBBcc, the number of heterozygous gene pairs (n) = 1
Type of gametes $=2^n = 2^1 = 2$

(129) Answer : (1)
Solution:
For flower colour, snapdragon plant shows incomplete dominance.
In F1 generation, 50 % would be Rr (Pink) and 50 % would be RR (Red). When these plants are crossed with the plant bearing white flowers, out of eight individuals, two produce white flowers, *i.e.*, 25%

(130) Answer : (2)
Hint:
In cymose inflorescence, flowers are borne in the basipetal succession.
Solution:
In racemose inflorescence, the shoot axis continue to grow indefinitely.

(131) Answer : (4)
Solution:
$1/5^{th}$ of the blood pumped by each ventricle in a minute is pumped out to kidneys
All the constituents of the plasma except proteins pass into the lumen.

(132) Answer : (3)
Solution:
Aldosterone is mineralocorticoid produced by the outermost layer of adrenal cortex *i.e.*, zona glomerulosa in response to decrease in the level of Na^+ in the blood plasma.

(133) Answer : (4)
Hint:
Thin filaments slide over thick filaments
Solution:
In a resting state, the edges of thin filaments on either side of the thick filaments partially overlap the free ends of the thick filaments leaving the central part of the thick filaments. This central part of thick filament, not overlapped by thin filaments is called the 'H' zone.

(134) Answer : (3)
Hint:
Bones of pelvic girdle
Solution:
• Based on location, three types of muscles are identified *i.e.*, skeletal, visceral and cardiac.
• Each middle ear contains three tiny bones – Malleus, incus, stapes
• Each coxal bone is formed by the fusion of three bones – Ilium, ischium, pubis
• Joints have been classified into three major structural forms – Fibrous, cartilaginous, synovial.

(135) Answer : (4)
Hint:
Controlled by medulla oblongata
Solution:
Brain controls the voluntary movements, balance of the body, functioning of vital involuntary organs (*eg*: lungs, heart, kidneys, *etc*.). Limbic system along with hypothalamus is involved in the regulation of sexual behaviour, expression of emotional reactions (*e.g,* excitement, pleasure, rage and fear), and motivation.
Respiratory rhythm centre is located in the medulla oblongata.

(136) Answer : (2)
Hint:
Cortisol and adrenaline are secreted under stress

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Solution:
The hormones produced during emotional stress are cortisol and catecholamines. Adrenaline increases metabolic rate and prepares the body for emergency situations

(137) Answer : (1)
Solution:
Neural organisation of *Hydra* contains apolar neuron.

(138) Answer : (3)
Solution:
In adult women, hypothyroidism may cause menstrual cycle to become irregular.

(139) Answer : (3)
Solution:
Parapodia are present in *Nereis*. Calcareous exoskeleton is seen in molluscs.

(140) Answer : (4)
Solution:
Coelomate animals.
The body cavity, which is lined by mesoderm is called coelom. Animals possessing coelom are called coelomate e.g. annelids, molluscs, arthropods, echinoderms, hemichordates and chordates. Water vascular system is a distinctive feature of echinoderms. The adult echinoderms are radially symmetrical animals. *Balanoglossus* and *Sepia* show bilateral symmetry.

(141) Answer : (4)
Solution:
Cyclostomes possess 6-15 pairs of gills.

(142) Answer : (4)
Solution:
The mouth parts of cockroach consist of a labrum (upper lip), a pair of mandibles, a pair of maxillae, a hypopharynx and a labium (lower lip).

(143) Answer : (2)
Solution:
In frogs, RBCs are nucleated.

(144) Answer : (3)
Solution:
Relaxin facilitates child birth by relaxing pubic symphysis.

(145) Answer : (3)
Solution:
Two meiotic divisions are required.

(146) Answer : (1)
Solution:
Prostatic fluid is thin and milky and alkaline in nature.

(147) Answer : (4)
Hint:
Secretion of Leydig cells helps in maintaining male accessory ducts and glands.
Solution:
The human male ejaculates about 200 to 300 million sperms during a coitus of which, for normal fertility, atleast 60% sperms must have normal shape and size and atleast 40% of them must show vigorous motility.

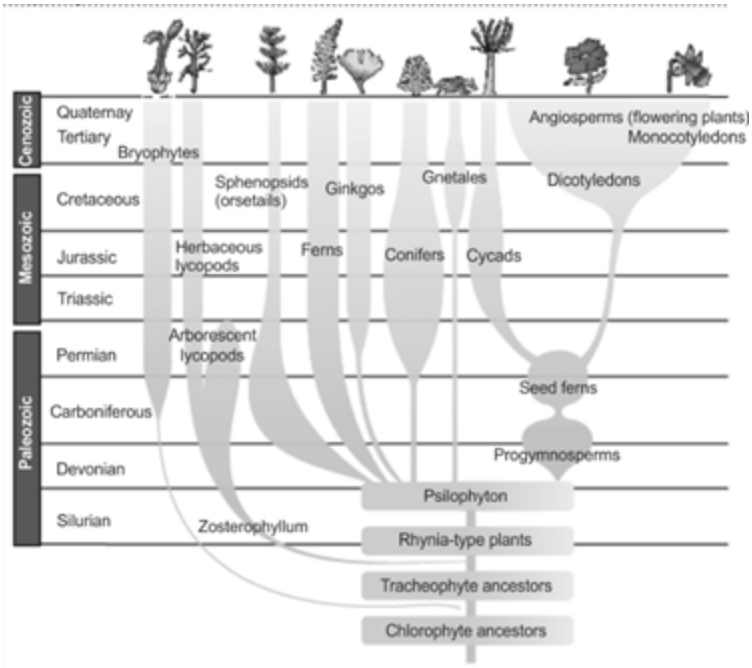
(148) Answer : (1)
Solution:
Morning after pills are also known as emergency pills. They may contain only progesterone, combination of progesterone and oestrogen or antiprogestin and prevent implantation.

(149) Answer : (3)
Solution:
Amniocentesis can be done in 20 weeks of pregnancy to know chromosomal defects.
Chorionic villi sampling can be done in 8 weeks of pregnancy.

(150) Answer : (4)
Solution:
Tubectomy is a sterilisation method for females.

(151) Answer : (3)
Solution:

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(152) Answer : (4)

Solution:

Mutations are random and directionless while Darwinian variations are small and directional.

(153) Answer : (3)

Solution:

Mucus coating of the epithelium, lining the respiratory and GI tract is a type of physical barrier.

(154) Answer : (2)

Solution:

Answer (2)

Abuse of testosterone brings aggressiveness in their behaviour.

(155) Answer : (3)

Hint:

Helper T-lymphocytes provide immunity to the body.

Solution:

In AIDS, due to decrease in the number of helper T-lymphocytes below a certain level (< 200 cells/mm^3), the person starts suffering from infections that could have been otherwise overcome such as those due to bacteria especially *Mycobacterium*, viruses, fungi and even parasites like *Toxoplasma*. The patient becomes so immunodeficient that he/she is unable to protect himself/herself against these infections.

(156) Answer : (4)

Solution:

In the given figure, both the cells are diploid and can produce gametes of four different genotypes.

(157) Answer : (4)

Solution:

RNA polymerase II catalyses the transcription of hnRNA. σ-factor facilitates the initiation of transcription in prokaryotes. Release factor facilitates the termination of translation. 2D structure of tRNA is clover leaf-like.

(158) Answer : (4)

Solution:

In the double helix structure of B-DNA, the pitch of the helix is approximately 3.4 nm (or 34 Å), and there are roughly 10 base pairs (bp) in each turn.

(159) Answer : (2)

Solution:

Structural genes that code for permease and transacetylase in *lac* operon are respectively *lac y* and *lac a*.

(160) Answer : (4)

Solution:

The plane of one base pair stacks over the other in double helix. This, in addition to H-bonds, confers stability of the helical structure of DNA.

(161) Answer : (3)

Hint:

Repetitive sequences are stretches of DNA sequences that are repeated many times.

Solution:

Polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting. Chromosome 1 has most genes (2968), and the Y has the fewest (231). Repetitive sequences are thought to have no direct coding functions, but they shed light on chromosome structure, dynamics and evolution.

(162) Answer : (4)

Solution:

Acetic acid is produced by *Acetobacter aceti* (bacterium).

(163) Answer : (4)

Solution:

Baculoviruses are pathogens that attack insects and other arthropods.
Trichoderma are very common in root ecosystems.

(164) Answer : (1)

Solution:

Abingdon tortoise in Galapagos Islands became extinct within a decade after goats were introduced on the island. The type of interaction between them is competition.

(165) Answer : (3)

Hint:

Predators help in maintaining species diversity in a community.

Solution:

Biological pest control methods used in agriculture are based on the ability of predator to regulate prey population.

(166) Answer : (1)

Solution:

There is always a loss of energy when it flows from a trophic level to the next trophic level. It is therefore, the pyramid of energy in an ecosystem is always upright.

(167) Answer : (3)

Solution:

Primary productivity is expressed in the terms of energy as (kcal m^{-2}) yr^{-1} and weight as (g m^{-2}) yr^{-1}.

(168) Answer : (3)

Solution:

Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds.

(169) Answer : (2)

Solution:

On a logarithmic scale the correct representation of species-area relationship is represented by the equation Log S = Log C + Z Log A

(170) Answer : (4)

Solution:

- Plants and fungi differ in their cell wall composition.
- All prokaryotic organisms were grouped together under Kingdom Monera.
- Mycoplasma are the smallest living cells known and they can survive without oxygen.

(171) Answer : (3)

Solution:

The *Lac*-Z gene codes for the β-galactosidase. The enzyme utilises its substrate to produce blue coloured product. In the given case, the non-transformants will form white coloured colonies whereas, transformants can be

- Recombinants (blue coloured colonies)
- Non-recombinants (white coloured colonies)

(172) Answer : (3)

Hint:

Complementary ends are required for linking

Solution:

Recombinant DNA can be created when the vector and the source DNA are cut with the same restriction enzyme.

(173) Answer : (1)

Solution:
In agarose gel electrophoresis, DNA fragments are stained with ethidium bromide and visualized under UV light.

(174) Answer : (3)
Solution:
α-1-antitrypsin is used to treat emphysema.

(175) Answer : (3)
Solution:
Probe may be a ssRNA or ssDNA.

(176) Answer : (4)
Solution:
A widely used diagnostic test for AIDS is ELISA.

(177) Answer : (3)
Hint:
RNAi is known as mRNA silencing.
Solution:
RNA interference is a novel strategy adopted to prevent the infestation of plants by nematodes.
RNAi is present in all eukaryotic organisms as a method of cellular defense.
RNAi involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA.

(178) Answer : (3)
Solution:
Post industrialisation, since the tree trunks became dark, dark winged moths were not too prominently visible to predators, hence they survived better.

(179) Answer : (1)
Solution:
During aestivation and hibernation gaseous exchange takes place through skin.

(180) Answer : (4)
Solution:
Presence of cilia on the inner surface of oviduct helps in the passage of ova and the movement shown is ciliary movement instead of flagellar movement.

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